Modeling Health Insurance Lapse Timing with Machine Learning and Survival Analysis

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Introduction

Problem | Dataset | Feature Engineering/Preprocessing

What is a Policy Lapse?

A lapse occurs when a health insurance policy ends because a customer stops paying premiums.

What drives lapse	Why insurers care about lapse rates
· Affordability pressures · Perceived value · Life changes · Engagement issues	· Revenue impact · Operational planning · Retention Strategy

Why Timing Matters in Lapse Prediction?

Lapse is not just whether a policy ends, it's *when* it ends

Time-to-lapse affects pricing, retention, and portfolio stability

Traditional classification models ignore censoring and timing whereas survival analysis handles it

DATASET OVERVIEW

- 215,428 policies after preprocessing
- **Event Rate:** 7.22% lapses, 92.78% censored
- **Duration:** mean = 23.5 years, range 0-86 years
- **Feature Groups**
  - Demographics (age, gender)
  - Premium & Claims Cost
  - Utilization (medical services)
  - Socioeconomic indicators (C_GI, C_II, etc.)
  - Policy Structure (type, distribution channel)
- **Missing Data:** Concentrated in socioeconomic variables; removed before modeling

Feature Engineering Pipeline

Age Groups:
7 behavioral bins (0-18, 19-25, … 65+)

Premium Engineering:
log-transform, deciles, centered premium

SES Tiers:
5-tier quantiles for 8 socioeconomic variables

Utilization Buckets:
U0-U4 based on medical service counts

Policy Structure:
one-hot encoding & premium interactions

Distribution Channels:
one-hot encoded (dist_A, dist_D, dist_I, etc.)

Reference Dummy Removal:
avoids multicollinearity for Cox

Scaling:
continuous variables standardized for Cox optimization

Dropped Raw Columns:
dates, raw SES, raw utilization, etc.

Train/Test/Validation Split:
60%/20%/20% with stratification

Data Leakage Features Removed:
Seniority_insured, seniority_policy

Survival Modeling Approach

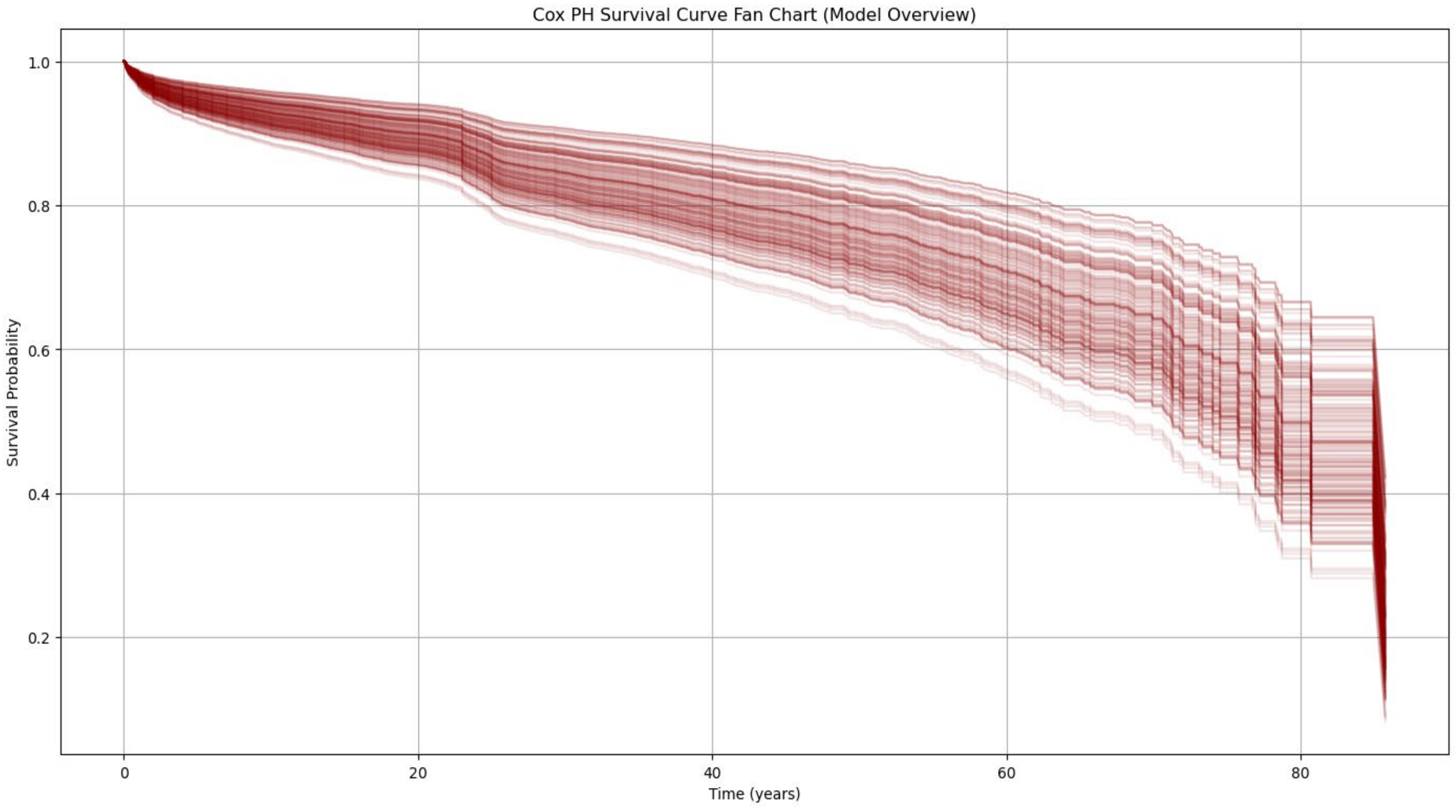
Model Selection | Cox Proportional Hazards | Random Survival Forests | Cox-Time

Model Selection

Classification	Model	PH Assumption	Nonlinear	Parameter Tuning	Interpretability
Traditional Statistical Method	CoxPH model	Yes	No	No	High
Ensemble Machine Learning	RSF	No	Yes	Yes	Moderate
Feedforward Deep Neural Network	CoxTime	No	Yes	Yes	Low

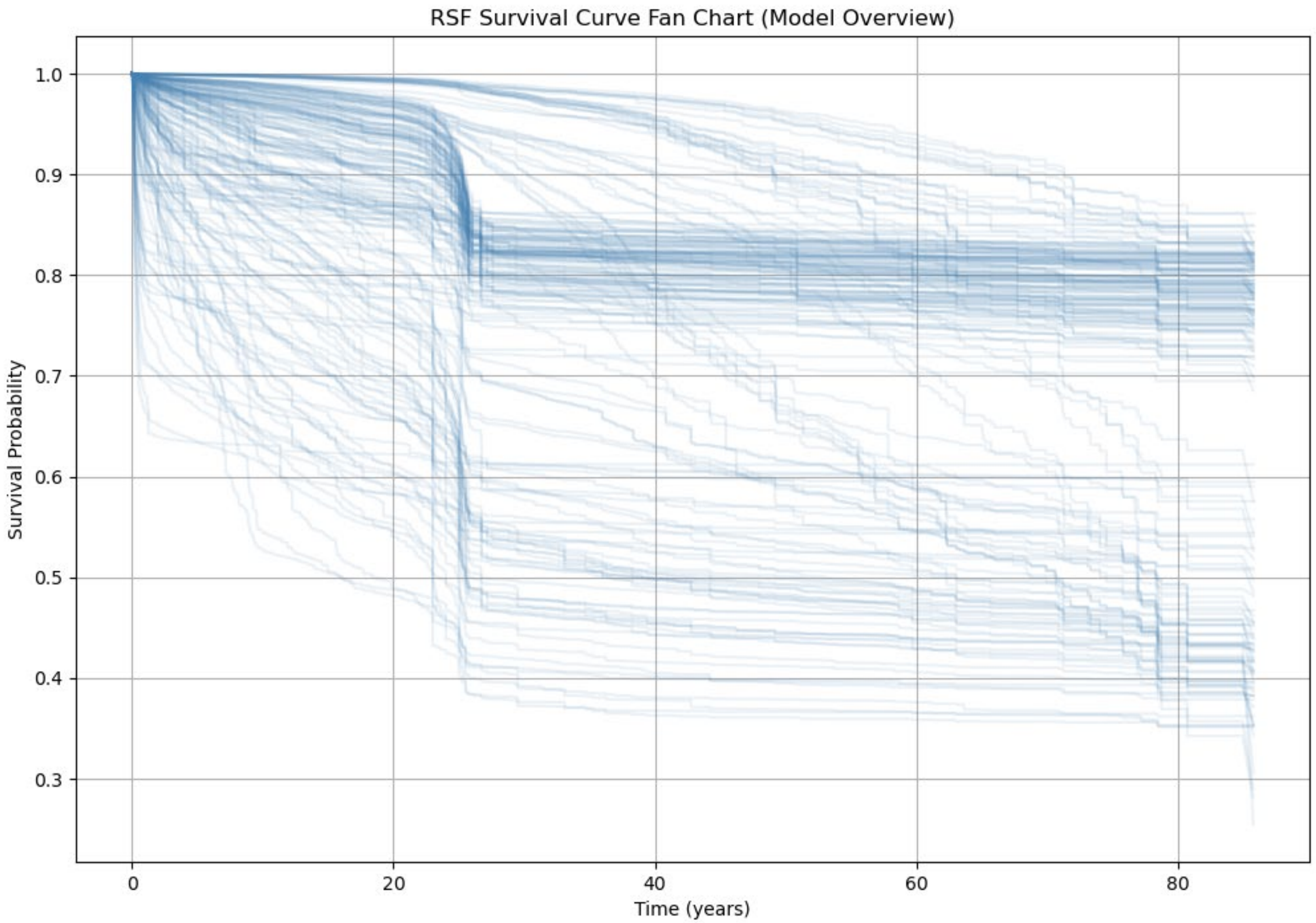
COX PROPORTIONAL HAZARDS MODEL

- Semi-Parametric model estimating hazard ratios
- Uses partial likelihood; handles censoring naturally
- PH assumption violated for several covariates



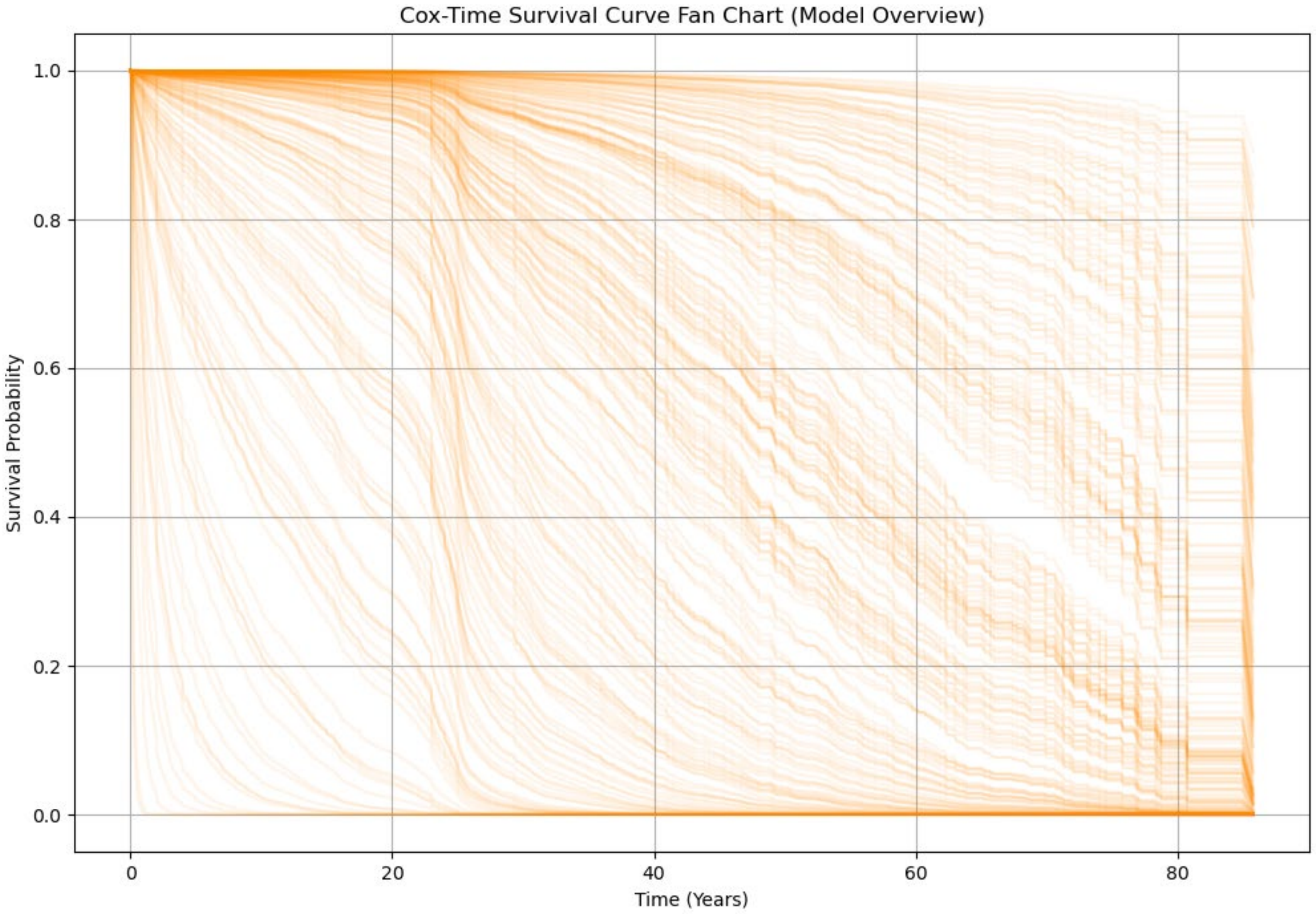
RANDOM SURVIVAL FOREST MODEL

- Ensemble of survival trees
- Captures nonlinearities & interactions automatically



COX-TIME SURVIVAL NEURAL NETWORK MODEL

- Time-dependent hazard modeling
- Captures nonlinear effects
- Flexible baseline hazard



Results

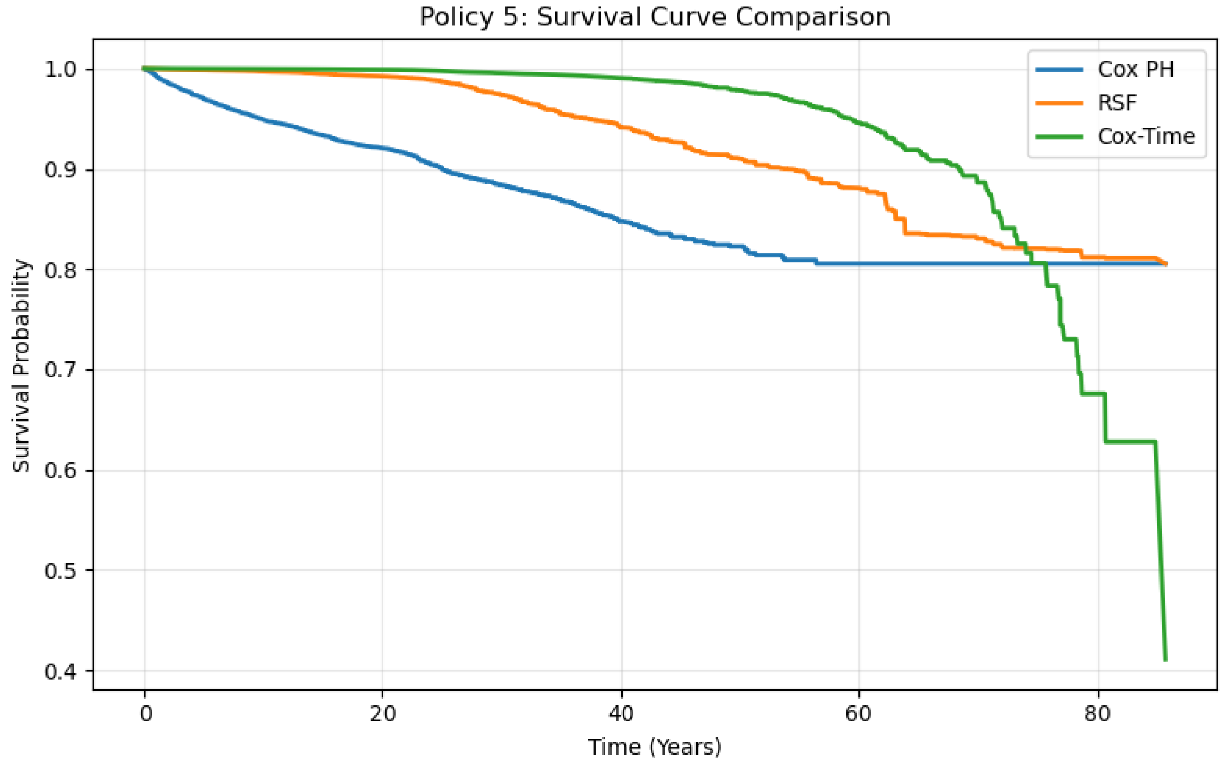
C-Index | Brier Score | Survival Curve Comparison | Feature Importance | Calibration Plot

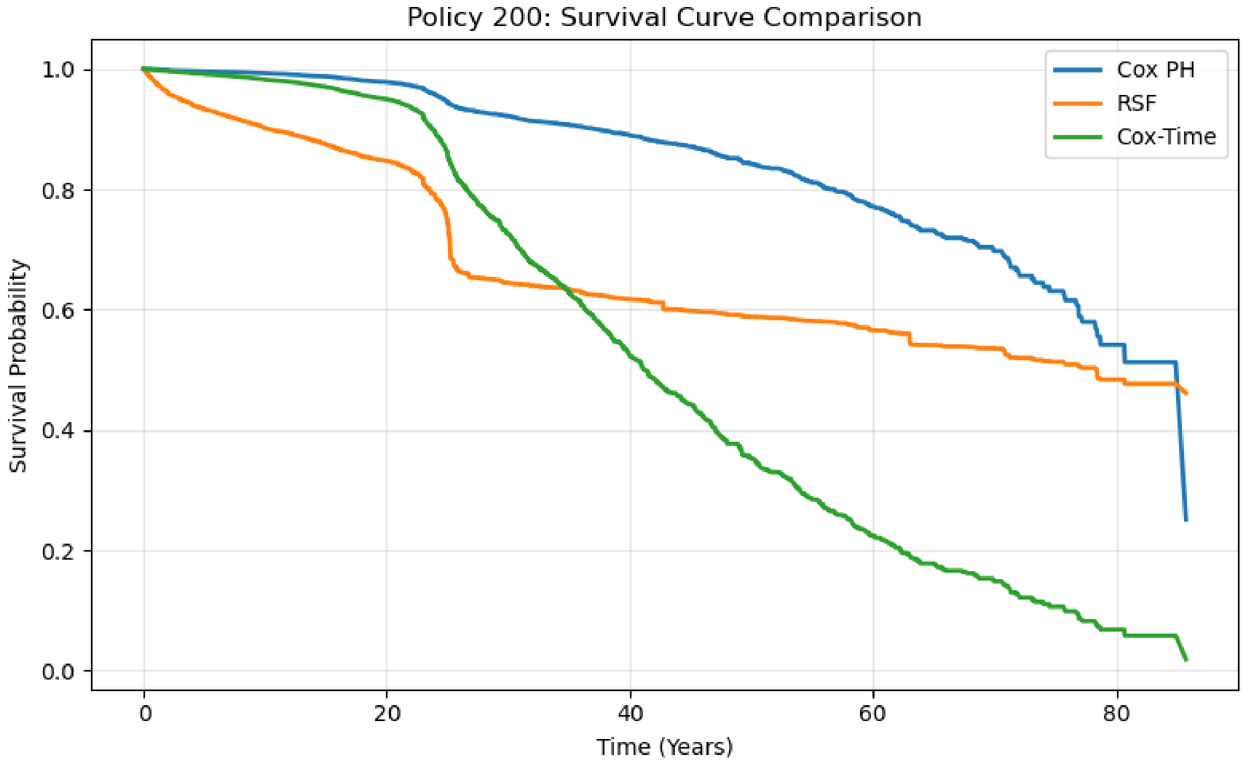
MODEL PERFORMANCE:
C-INDEX & BRIER SCORES

Brier Scores		
	Mean	Integrated
CPH	0.1305	0.1298
RSF	0.1039	0.1023
Cox-Time	0.0396	0.0784

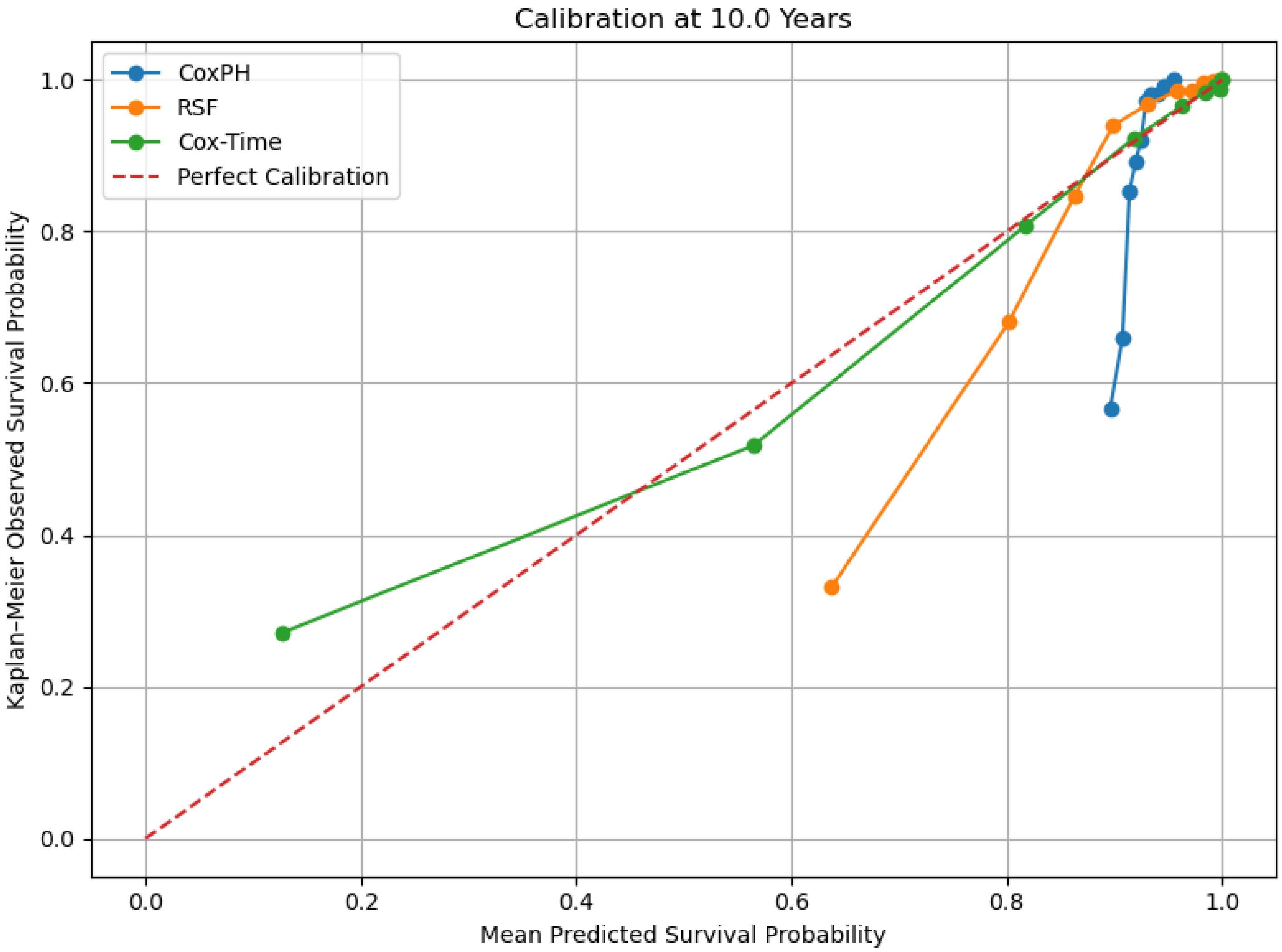
C-Index Scores			
	Train	Test	Validation
CPH	0.8358	0.8353	0.8340
RSF	0.9069	0.9015	0.8978
Cox-Time	0.9167	0.9177	0.9127

Survival Curve Comparison (Policy 5 & 200)

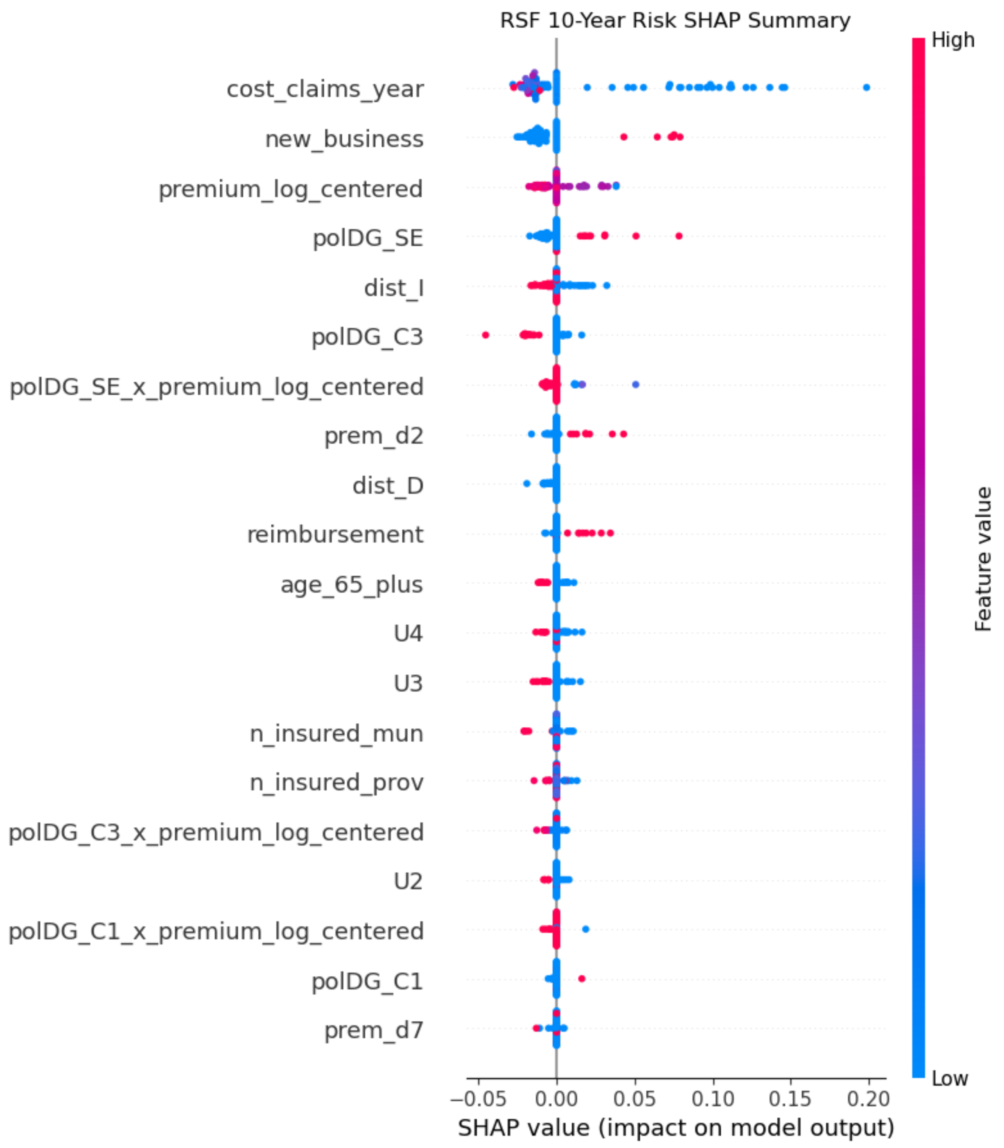


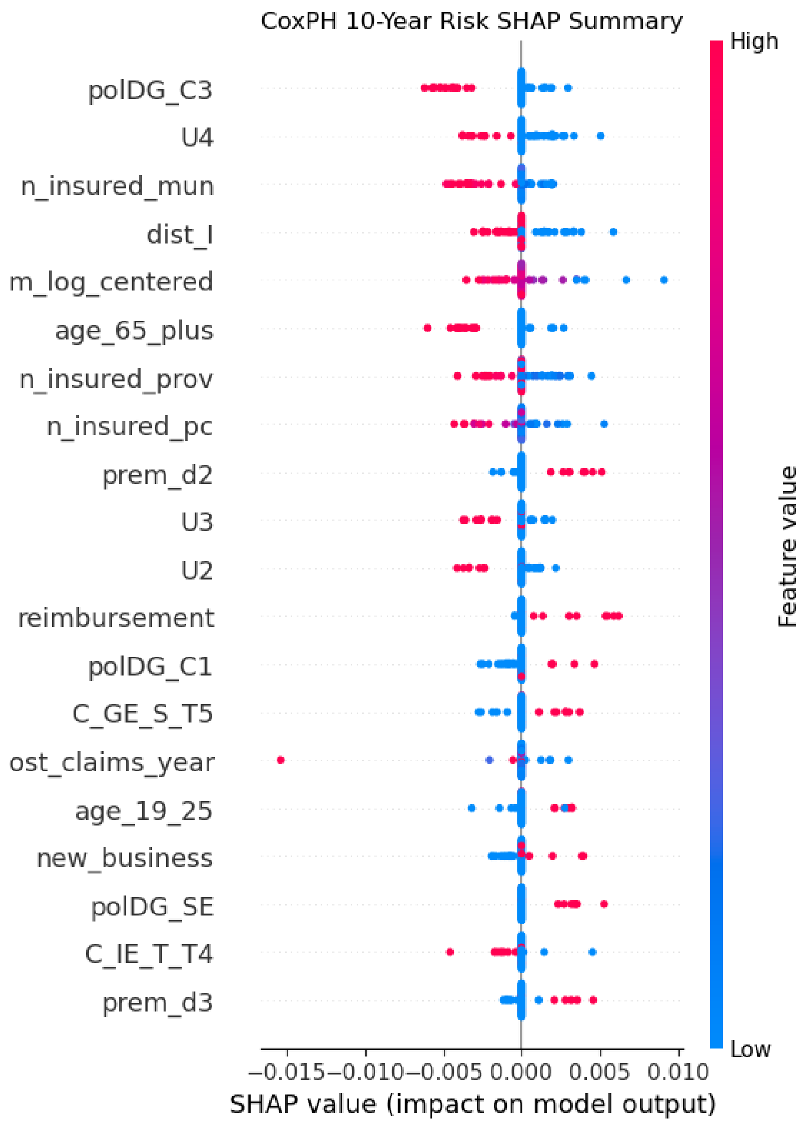


CALIBRATION PLOT

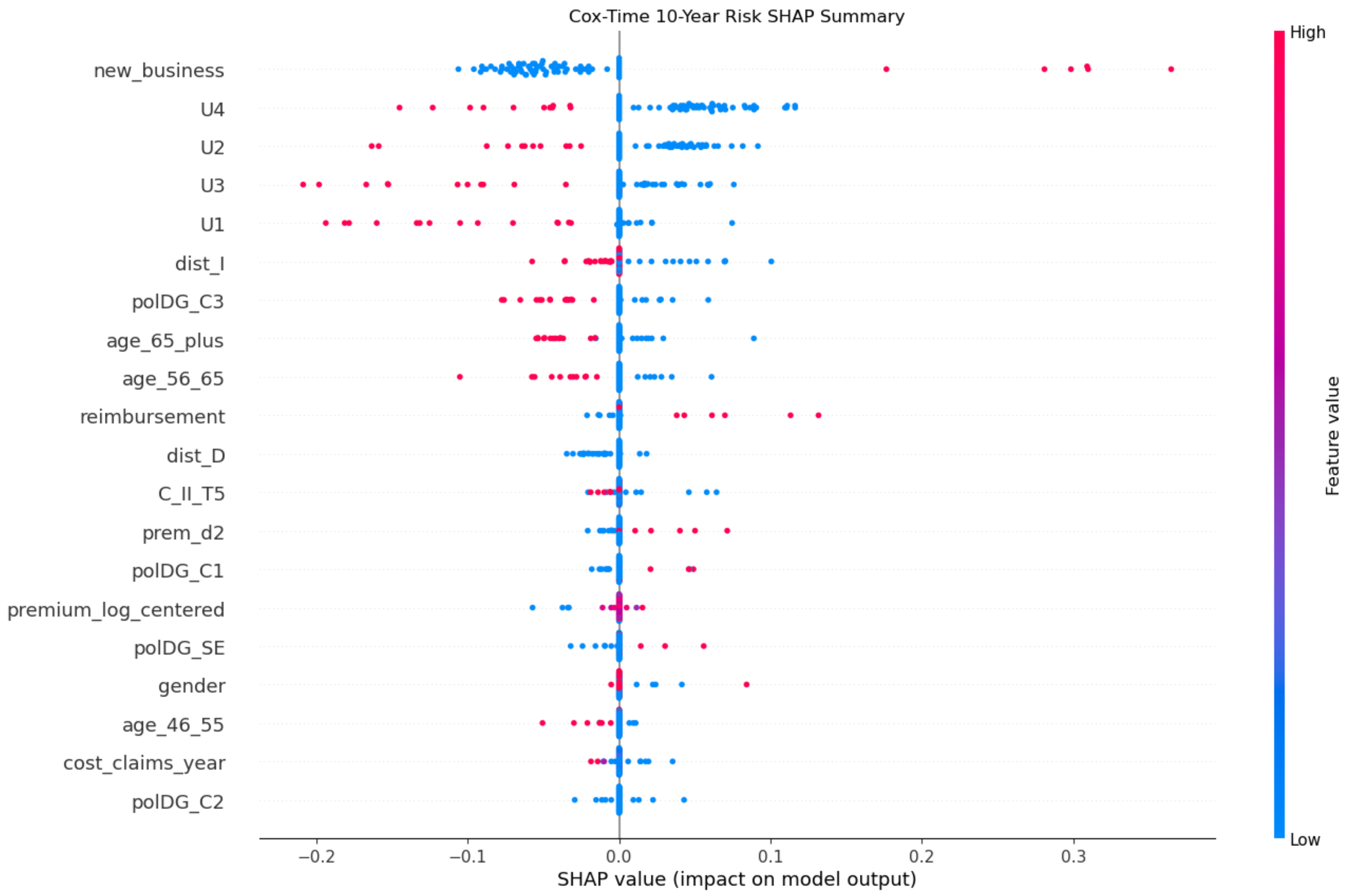


What Drives Lapse Risk?





What Drives Lapse Risk? (cont)



Future Work

FUTURE WORK

Richer Data Sources: incorporate multi-year policy histories, behavioral indicators, and external socioeconomic features to strengthen lapse signal detection

Fairness and Stability Analysis: test group-specific calibration, demographic fairness metrics, and bias mitigation strategies

Temporal Modeling of Lapse Behavior: incorporate recurrent or sequence-based models to capture evolving policyholder behavior and time-dependent risk

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Thank You

Questions?